

PGR Platform

System Architecture & Deployment Sizing

Approximation for real-institution deployment. Derived from the current codebase, 224 passing tests, verified read p95 6 ms, dashboards 9–43 ms, cohort integrity 0.17 s / 266 students.

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1 · Topology

Modular monolith. Three runnable processes plus Postgres, optional read replica, object store, mail relay, and an HTTPS terminator.

Tier	What it is	Process / service
Edge	TLS + WAF + rate limits	nginx / CloudFront / Traefik
Frontend	Next.js 14 SSR + static	node (npm run start)
API	FastAPI (uvicorn --workers N)	python -m uvicorn app.main:app
Worker	APScheduler: outbox, jobs, notifs	python -m app.worker
Database	PostgreSQL 18 primary	managed or self-hosted
Read replica	Postgres physical replica	wired via DATABASE_REPLICA_URL
Object store	Documents (theses, letters, IDs)	S3 / MinIO — swap STORAGE_BACKEND
Mail	Transactional email	SES / Mailgun / SMTP relay
Secrets	APP_SECRET_KEY, DB pw, HMAC keys	AWS SM / Vault / Azure KV
Observability	Traces, metrics, logs, errors	OTel → Grafana/DD; Sentry
Backup	PITR + off-site archive	pgBackRest / Barman / managed

2 · Capacity bands

Pick the row that matches active PGR population. Concurrency estimate assumes ~10 % of the population is active in the app at peak, plus office staff.

Band	Active PGRs	Peak concurrent users	Requests / min	DB size (5 yr)
Small	≤ 500	≤ 50	≤ 600	≤ 20 GB
Medium	500–2,000	50–200	600–2,500	20–80 GB
Large	2,000–5,000	200–500	2.5–6 k	80–200 GB
X-Large	> 5,000	500+	6 k+	200 GB+

3 · VM / container specs per band

Sized from measured behaviour: request handler p95 ~6 ms → 1 vCPU sustains ~150 req/s at 40 % headroom; add 1 vCPU per 250 concurrent users. Worker is IO-bound. Postgres sized for working set in RAM at each band.

3.1 · Application tier

Component	Small	Medium	Large	X-Large
Edge / TLS	shared / managed	shared / managed	shared / managed	shared / managed
Frontend (Next.js)	1× 2 vCPU / 2 GB	2× 2 vCPU / 2 GB	3× 2 vCPU / 4 GB	4× 4 vCPU / 4 GB
API (FastAPI)	1× 2 vCPU / 4 GB 2 workers	2× 2 vCPU / 4 GB 2 workers ea	3× 4 vCPU / 8 GB 4 workers ea	6× 4 vCPU / 8 GB 4 workers ea
Worker	1× 2 vCPU / 4 GB	1× 2 vCPU / 4 GB	2× 2 vCPU / 4 GB	2× 4 vCPU / 8 GB
Storage per node	30 GB SSD	30 GB SSD	40 GB SSD	50 GB SSD

3.2 · Data tier

Component	Small	Medium	Large	X-Large
Postgres primary	2 vCPU / 8 GB / 100 GB SSD	3× 2 vCPU / 16 GB / 200 GB SSD	8× 2 vCPU / 32 GB / 500 GB SSD	15× 2 vCPU / 64 GB / 1 TB SSD
Postgres replica	optional	1× same as primary	1–2× same as primary	2× same as primary
PgBouncer	colocate on primary	1× 1 vCPU / 1 GB	2× 1 vCPU / 1 GB	3× 2 vCPU / 2 GB
Object store	S3 bucket, 20 GB	S3 bucket, 100 GB	S3 bucket, 500 GB	S3 bucket, 2 TB
Backup (PITR)	daily + WAL, 30-day RPO	daily + WAL, 30-day	hourly + WAL, 30-day	hourly + WAL, 90-day

4 · Cloud SKU cheat-sheet (Medium band)

Role	AWS	Azure	GCP	On-prem VM
Frontend	t3.small × 2	B2s × 2	e2-small × 2	2 vCPU / 2 GB × 2
API	t3.medium × 2	B2ms × 2	e2-medium × 2	2 vCPU / 4 GB × 2
Worker	t3.medium × 1	B2ms × 1	e2-medium × 1	2 vCPU / 4 GB × 1
Postgres	RDS db.m6g.large 16 GB, gp3 200 GB	Flexible D4s v3 200 GB P10	Cloud SQL 4 vCPU 16 GB, 200 GB PD-SSD	4 vCPU / 16 GB / 200 GB SSD
Objects	S3 std	Blob Std LRS	GCS std	MinIO / NAS
Mail	SES	ACS / SendGrid	SendGrid	SMTP relay

5 · Network & security

Rule	Value
Public ingress	443/TCP only, HTTP→HTTPS redirect at edge
Internal API	8000/TCP, private VPC, closed to public
DB port	5432/TCP, only from API + worker SG
Object store	IAM role on API + worker; presigned URLs to browser
Rate limit — /auth/*	10 / min / IP + 20 / hour / account
Rate limit — /assistant	30 / min / user
Rate limit — inbound webhooks	200 / min / partner sourceId prefix
mTLS or IP allow-list	Inbound partner webhooks (Research / HR / Finance)
WAF baseline	OWASP CRS, log-only for 2 weeks then enforce
Egress	SMTP relay + partner webhook targets only
Secrets rotation	APP_SECRET_KEY 90 d, DB pw 180 d, HMAC per partner 180 d

6 · Storage growth

Data class	Per active PGR / year	Retention	Where
Structured records	≈ 60 KB	Life of record + policy	Postgres
Audit rows	≈ 40 KB	≥ 7 years	Postgres
Documents (thesis, letters, scans)	≈ 100 MB	Life + 6 years	Object store
Notifications archive	≈ 10 KB	≥ 2 years	Postgres
Log / trace / metric	≈ 200 MB	30–90 d hot, 1 y warm	Log store
Backups (PITR + off-site)	≈ 2 × DB size	30–90 d	Object store

7 · SLOs and how they are met

SLO	Target	How it is achieved
API read p95	< 150 ms	JWT-claim principal (no DB round-trip); replica routing; index coverage
Dashboard tile p95	< 300 ms	Materialised 360 views refreshed by worker (planned)
Login p95	< 400 ms	pbkdf2 cost tuned; rate-limit before hash
Availability	99.5 %	2× API, 2× frontend behind LB; managed Postgres
RPO	≤ 15 min	WAL shipping
RTO	≤ 4 h	Restore from PITR + replay outbox from source of truth
Outbox delivery	at-least-once, ≤ 60 s	Worker 20 s dispatch cycle + retry/backoff

8 · Scale-out rules

Signal	Action
API CPU > 60 % for 10 min	+1 API pod (horizontal)
API p95 latency > 2× baseline	+1 API pod, check DB slow-query log
Worker queue depth (outbox pending) > 500	+1 worker (partition by event type prefix)
Postgres CPU > 70 % sustained	Route reads to replica; add replica if needed
Postgres storage > 75 %	Grow volume (managed Postgres) or archive audit rows
Notification lag > 2 min	+1 worker or split notification channel

9 · Runtime dependencies

Kind	Version	Notes
Python	3.12+	3.13 is the tested dev runtime
PostgreSQL	16+ (18 in dev)	Enums used; upgrades tested via Alembic
Node.js	20 LTS+	Frontend build + start
OS	Linux (Ubuntu 22.04+)	Windows is dev-only
Container base	python:3.12-slim, node:20-alpine	Dockerfile in backend/; verify build in CI
TLS certs	Managed (ACM / cert-manager)	Renewal automated

10 · Environment plan

Env	Purpose	Sizing	Data
dev	engineer laptops	docker compose	seed script
ci	PR test runs	ephemeral, 2 vCPU / 4 GB	SQLite in-memory (as today)
staging	integration + UAT	Small band	sanitised prod snapshot, monthly
prod	live	band matching population	live
dr	disaster recovery	cold replica of prod	restored from PITR nightly

11 · Integration endpoints (must be provisioned)

Partner	Direction	Endpoint	Auth
Research Office	inbound	POST /api/v1/integration/inbound	HMAC-SHA256 (X-Signature)
HR	inbound	POST /api/v1/integration/inbound	HMAC-SHA256
Finance	outbound	OUTBOX_TARGET_URL (Finance)	HMAC + mTLS recommended
Finance	inbound	POST /api/v1/integration/inbound (handler TBD)	HMAC
SSO (later)	browser	OIDC via edge	Institutional IdP

12 · Rough monthly infra cost (Medium band, single region)

Order-of-magnitude only. Assumes managed services, 3-yr reservation where applicable, no support tier included, GBP.

Line	AWS	Azure	GCP
Compute (2 API + 2 FE + 1 worker)	£220	£240	£210
Managed Postgres 4 vCPU / 16 GB	£280	£300	£260
Read replica	£280	£300	£260
Object storage (100 GB + egress)	£15	£15	£15
Backups (PITR + snapshots)	£40	£40	£35
Load balancer + WAF	£45	£45	£40
Log / metric / trace (moderate volume)	£120	£130	£110
Email (SES / equivalent, 100k/mo)	£10	£15	£15
Total (approx)	£1.0 k	£1.1 k	£0.95 k

13 · Deployment gates before first go-live

#	Gate	Owner
1	TLS terminator with valid cert + HSTS	Platform / SRE
2	Rate limits on /auth/* and /integration/inbound	Platform
3	APP_SECRET_KEY + HMAC keys in secret store	Security
4	Postgres backup + tested restore drill	DBA
5	Observability wired (OTel + Sentry + logs)	Platform
6	Subject-access + erasure endpoints live	Product + IG
7	Statutory profile signed off by Registry / HESA	Registry
8	Integration contracts signed with Research / HR / Finance	Programme
9	Rollback runbook exercised end-to-end	SRE
10	Load test at target band passes SLOs	Platform

Read alongside `PGR_Platform_Critical_Gaps.pdf` — several rows in §13 are the same items called out as gaps to close before institution go-live.